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Anopheline fauna and malaria transmission in four ecologically distinct zones in Cameroon

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ABSTRACT

Knowledge of baseline malaria transmission intensity in a given environment is important to guide malaria control interventions. In Cameroon, recent information on malaria transmission intensity is insufficient. Therefore, an entomological study was conducted in four ecologically different sites throughout the country to assess the seasonal patterns in malaria transmission intensity. *Anopheles arabiensis* was the main vector in six of the nine study sites, while *An. gambiae* sensu stricto was the most important vector in the other three sites. Clear differences in entomological inoculation rates (EIR) were observed between the study sites, ranging from 0.1 infective bites per person per night in the sahelian zone of the country to 5.5 infective bites per person per night in the forest zone. Based on the observed behaviour of the vectors, insecticide-treated bed nets will be highly effective in controlling malaria. However, in the high transmission areas, additional measures will be needed to reduce the malaria burden to acceptable levels.

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1. Introduction

Malaria remains one of the most serious vector borne diseases in the world and is highly endemic throughout Cameroon (Greenwood and Mutabingwa, 2002; Quakyi et al., 2000). The World Health Organisation estimates that malaria affects approximately 300–500 million people, killing 1.5–2.5 million each year. More than 90% of these cases are reported in sub-Saharan Africa with more than 70% of inhabitants in certain areas of this region being chronically infected by *Plasmodium falciparum* (OMS, 1998). In Cameroon, nearly 10,000 clinical cases, responsible for 40–45% hospital consultations, 20% hospitalizations and 35–40% deaths are recorded each year. The most affected are children under the age of five and pregnant women (WHO/CAMINFOS, 2002). Despite effort expended by the National Malaria Control Programme to curb the disease burden, its endemicity appears not to be changing much. This is thought to be due to the development of drug resistance in the parasite and insecticide resistance in the mosquito vectors. Studies have attributed observation to the presence of very efficient mosquito vectors of *P. falciparum* whose distribution and bionomics in transmission vary greatly across the different geo-climatic zones of the country (Atangana et al., 2009; Bigoga et al., 2007; Cavalié and Mouchet, 1961; Fondjo, 1996; Tchuinkam, 1994).

Of the at least 48 malaria vector species documented from Cameroon, *An. gambiae* s.s., *An. arabiensis* and *An. funestus* are widespread and constitute the principal malaria vectors. *An. arabiensis* is restricted to the northern sudano-savannah and sahelian parts of the country and *An. melas* to the coastal zones. *An. hancocki*, *An. nili* and *An. moucheti* play a secondary role in transmission in the forest areas, and their relative importance may vary from one bio-ecological zone to another. Thus, knowledge of malaria transmission and its evolution over time is extremely important for selecting and targeting malaria control interventions (Beier et al., 1999). Indeed, protective efficacy of vector control will be dependent on the initial intensity of malaria transmission (Lengeler et al., 1997). Cameroon is characterized by several bio-geographical zones and the National Malaria Control Programme under the Roll Back Malaria initiative urgently needs up-to-date information on the dynamics of malaria transmission from every part of the coun-

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try. This paper reports on work carried out between 2003 and 2007 to characterize the anopheline fauna and their contribution to malaria transmission in nine locations; six in the North, two in the western highlands and one in the southern forested zone within the four bio-ecological zones in Cameroon.

2. Materials and methods

2.1. Study site description and ethical consideration

Mosquitoes were collected from nine localities as follows: Mokolo (13°80′N; 10°75′E; 900 m a.s.l.), Godola (10°70′N; 14°25′E; 480 m a.s.l.), Gounougou (9°05′N; 13°73′E; 194 m a.s.l.), Djippordé (9°02′N; 13°39′E) and Mayo Mbocki (8°43′N; 13°33′E; 298 m a.s.l.) in the northern sudano-sahelian savannah zone, Vourgné-Sanfou (7°50′ N; 13°43′E) in the humid savannah zone of the Adamaoua region, Essong (2°31′N; 12°39′E) in the southern forest zone, Mangoum (5°28′N; 10°33′E; 1100 m a.s.l.) and Ngwatta (05°18′ N; 09°54′E; 700 m a.s.l.) in the western highlands zone.

The northern sudano-sahelian savannah zone is characterized by two seasons; one long dry season (November–May) and a short rainy season (June–October), with a mean annual temperature of 28 °C and mean annual rainfall ranging from 400 to 1000 mm. The Adamaoua region (humid-savannah highland area) has an altitudinal climate that differs from that of the northern savannah zone. It has the mean temperature of 21 °C and mean annual rainfall of 1500 mm. The climate in the southern forest region has four distinct seasons: the long dry season (December–February), the long rainy season (September–November), the short dry season (July–August) and the short rainy season (March–June), with the mean annual rainfall of 1500–2000 mm and mean annual temperature of 25 °C. The climate in the western highlands is characterized by one long rainy season (March–October), with high annual rainfall (1800–2500 mm) and mean annual temperature below 22 °C.

Table 1 gives the predominant land cover in the various sampling sites and the period (month, year and season) during which mosquitoes were found, mainly in the northern savannah, Adamaoua and western highland regions. Apart from Vourgné Sanfou and Ngwatta where surveys were conducted only during the rainy season, all surveys were conducted during the dry and rainy seasons.

An ethics clearance document for this study was obtained from the National Ethics Review Committee of Cameroon. Participation in mosquito collection was strictly voluntary. All collectors were clearly explained the purpose of the study and any risks involved, following which they were trained on the collection method prior to commencement of collections. All collectors were placed on chemoprophylaxis throughout the study period according to recommendations of the National Malaria Control Programme.

2.2. Field collection and processing of adult anopheline specimens

For each study site, except in Vourgné sanfou and Ngwatta, two entomological surveys were conducted between August 2003 and August 2007. For three consecutive nights (6:00 pm–6:00 am) and over a six-day period during each survey at each site (36 man nights per site), mosquitoes were captured indoors on volunteer collectors using the landing catch method. All volunteers gave free and informed consent for capturing mosquitoes and were given the appropriate training and appropriate malaria prophylaxis following the guidelines of the National Malaria Control Programme throughout the study duration. Anopheline and culicine mosquitoes were sorted and the female anophelines morphologically identified (Gillies and De Meillon, 1968; Gillies and Coetzee, 1987). Individual mosquitoes were preserved on cotton wool placed over a desiccant (silica gel) in labeled tubes and transported to the laboratory for further immunological and molecular biology analyses.

2.3. Laboratory processing of anophelines

In sites where the number of *An. gambiae* mosquitoes collected per survey exceeded 50, the legs and wings of at least 50 were used to identify the siblings by PCR (Scott et al., 1993). Positive control mosquitoes for *An. melas*, *An. gambiae* s.s., *An. arabiensis* and *An. quadriannulatus* and negative controls (tubes containing all the components of a PCR reaction mixture but without template DNA) were included. Analysis of *An. gambiae* s.s. molecular forms (M and S) in a subset of individuals was done using a PCR-RFLP method previously described by Favia et al. (2001).

The head-thorax portion of each mosquito was separated from the rest of the body, homogenized in blocking buffer (0.5% Casein, 0.1 N NaOH, 1× PBS) and assayed for the presence of *P. falciparum* circumsporozoite protein (CSP) by ELISA (Burkot et al., 1984; Wirtz et al., 1987). Positive controls (Kikergaard & Perry Laboratories, USA) and negative controls (uninfected laboratory reared mosquitoes) were assayed simultaneously. A specimen was considered positive if a visual green colour was detected with an OD value (at 405 nm) of at least the mean of the negative controls plus three standard deviations. This was to minimize having false positive CSP ELISA results.

2.4. Data analyses

Man biting rate (ma) was calculated as the average number of bites received per person per night of collection; infection rate was measured as the proportion of mosquitoes found to contain circumsporozoite protein by ELISA; entomological inoculation rate (EIR) was calculated as the product of the man biting rate and the infection rate. Tests for differences in proportions of categorical variables, including species and CSP indices, were performed by chi-square analysis. All tests were performed at the 5% significance level.

3. Results

3.1. Anopheles species composition and abundance

Table 2 shows the relative abundance of the vector species and their distribution in the different locations. A total of 4629 anophelines was collected throughout the study. The species caught were: *An. gambiae* s.s., *An. arabiensis*, *An. funestus*, *An. rufipes*, *An. pharoensis*, *An. nili*, *An. squamosus*, *An. coustani*, *An. paludis*, *An. moucheti* and *An. ziemanni*. Most of these species (73.6% of the anopheline fauna) were collected during the rainy season, 88.1% of the rainy season collections being *An. gambiae* s.s. *An. arabiensis* was the predominant vector in the six northern site and distributed as follows: Godola (60.7%, $n = 89$), Mokolo (90.3%, $n = 31$), Djippordé (75.4%, $n = 61$), Gounougou (51%, $n = 443$), Mayo Mbocki (51.3%, $n = 437$) and Vourgné sanfou (74.1%, $n = 143$). On the other hand, *An. gambiae* s.s. was the dominant species in Essong (99.6%, $n = 1913$); while in Mokolo it represented only 5% of the total *Anopheles* collected as it is not better adapted to arid zones. Compared to Mangoum, Essong, Mayo Mbocki and Gounougou, anopheline collections from Godola, Mokolo and Djippordé were lower both in the dry and in the rainy seasons (Table 2).

3.2. Biting rates and seasonality

The average human biting rate (ma) varied temporarily. However, this was dependent on the period of collection per site

Table 1
Description of the main land cover and study period in the collection sites.

Region	Locality	Land covers	Month and year	Season
Northern savannah	Mokolo	Urban area	August 2003	Rainy
			November 2004	Dry
	Godola	Farmland	August 2003	Rainy
			November 2004	Dry
	Gounougou	Rice cultivation	June 2006	Rainy
			October 2006	Dry
	Djippordé	Urban area	June 2006	Rainy
			October 2006	Dry
	Mayo Mbocki	Farmland	June 2006	Rainy
			October 2006	Dry
	Vourgne Sanfou	Farmland	August 2007	Rainy
Southern forested	Essong	Farmland and fish pond	August 2005	Rainy
			May 2006	
Western highlands	Mangoum	Farmland	October 2005	Dry
			May 2006	Rainy
	Ngwatta	Farmland	May 2006	

(Table 2). In the rural areas of Essong and Mangoum situated in the southern forested and western highland regions, the biting rate was high and varied between 28.3 and 56.8 bites per person per night (bpn) during the dry and wet seasons respectively, and was mainly due to *An. gambiae* s.s. In the northern savannah zone, there were 9 and 8 bpn due to *An. arabiensis* in Mayo Mbocki and Gounougou respectively during the rainy season. In the western highlands region, the bite rate was lower in Ngwatta during the rainy season (16 bpn) whereas in the northern savannah zone, the bite rates were lower in Godola (1.8 bpn) and Mokolo (1.5 bpn). Generally, where mosquitoes were collected during both the dry and the rainy seasons, the bite rates were lower during the dry season and the difference was statistically significant ($p < 0.05$). However, the higher bite rates due to *An. gambiae* s.s. and *An. arabiensis* were observed during the dry season in Essong (48 bpn), in the southern forested zone and in Gounougou (5 bpn) in the northern savannah zone respectively.

3.3. Infection rates

In all, 2025 female anopheline specimens belonging to 10 species collected in the different study sites were processed for infectivity by ELISA. Of these, ninety-one *Anopheles* including *An. gambiae* (2.0%), *An. arabiensis* (2.63%), *An. funestus* (0.75%), *An. pharoensis* (3.57%) and *An. coustani* (1.47%) were positive for *P. falciparum* circumsporozoite proteins (Table 3). Generally, the difference in circumsporozoite protein indices between the anopheline species in the different sites was not significant ($\chi^2 = 10.4$; d.f. = 4; $p = 0.3$). However, the observed difference between *An. arabiensis* and *An. gambiae* was not significant ($\chi^2 = 0.98$; d.f. = 1; $p = 0.46$).

3.4. Entomological inoculation rates (EIR)

The highest number of infective bites per person per night (ib/p/n) was recorded with *An. gambiae* s.s. in Mangoum, during the rainy season (5.0 ib/p/n). *An. gambiae* s.s. was also responsible for 100% of the transmission in Essong, Mangoum and Ngwatta, whereas *An. arabiensis* was the main vector in Mayo Mbocki and responsible for 80% of the transmission. The infection rates and EIR varied by season and study site. The lowest values were recorded in Vourgné Sanfou, Godola, Djippordé and Gounougou during the rainy season. No transmission was observed in the study sites situated in the northern savannah zone during the dry season (Table 3). Overall, mean EIR of 9.66 and 1.47 ib/m/n were observed during the rainy and dry seasons respectively. Thus, an individual in any of the sites would on the average receive 10 infective mosquito bites every night during the rainy season and 2 infective bites every night during the dry season.

4. Discussion

Vector control is an indispensable aspect of the global malaria control strategy. However, the success of malaria vector control requires accurate and up-to-date information on the bionomics of the species involved in disease transmission. The National Malaria Control Programme has in the recent past reinforced malaria control through the wide-scale distribution long lasting insecticide treated bed nets, particularly to the most affected, children less than five years and pregnant women. Nevertheless, a lot more remains to be done as the paradigms of malaria transmission in many areas of Cameroon are still poorly understood. Thus, the current study adds to existing data, some useful information on the role of Anophelines in malaria transmission that would contribute to better vector intervention in the areas studied.

During this study from August 2003 to August 2007, eleven (11) anopheline species out of the 48 already recorded in Cameroon were collected. This observation corroborates those of previous studies reviewing the anopheline fauna of Cameroon (Antonio-Nkondjio et al., 2006; Bigoga et al., 2007; Brunhes et al., 2003; Cohuet et al., 2000; Henry et al., 2003; Hervy et al., 1998). The plethora of anopheline populations' species reflects the diversity of the biotopes in the country. However, only *An. gambiae*, *An. arabiensis*, *An. funestus*, *An. pharoensis* and *An. coustani* were found to be susceptible to *P. falciparum* infection. Infections in *An. pharoensis* and *An. coustani* have been reported in several studies in Cameroon (Antonio-Nkondjio et al., 2006; Robert et al., 1992) and these species play only a secondary role in transmission. Thus, despite their contribution to malaria transmission in some of the surveyed areas it is unlikely that these species could maintain stable parasite transmission on their own in the absence of the major vectors in the studied areas.

The intensity of malaria transmission in the northern parts of the country was lower in the peri-urban Godola village and in the urban Mokolo and Djippordé areas. This was probably due to the low number of anopheles species collected in these areas. In Gounougou, a rice cultivation area, malaria transmission was observably lower, despite the high anopheline density. *An. arabiensis* and *An. funestus* were the main malaria vectors. This observation is consistent with the distribution of these species in Cameroon and in the arid savannah areas of West Africa (Antonio-Nkondjio et al., 2008; Dia et al., 2003; Dolo et al., 2004; Doussou-yovo et al., 1995; Koudou

Table 2
Composition of the anopheline fauna and average human biting rate (HBR) in the different study sites by season (numbers in parentheses represents the average biting rate per vector species study site).

Anopheles species	Season	Northern region						Southern forested region	Western highland region		Total
		Godola	Mokolo	Djippordé	Gounougou	Mayo Mbocki	Vourgné Sanfou	Essong	Mangoum	Ngwatta	
An. Arabiensis	Dry	21 (0.58)	0	0	93 (2.6)	64 (1.8)	0	0	0	0	178 (4.9)
	Rainy	33 (0.92)	28 (0.78)	46 (1.28)	133 (3.7)	160 (4.4)	106 (2.9)	0	0	0	506 (14)
An. gambiae s.s.	Dry	0	0	0	0	3 (0.2)	0	872 (24.2)	147 (4.1)	0	1023 (28.4)
	Rainy	0	0	0	1 (0.03)	29 (0.8)	75 (21)	1034 (28.7)	1012 (28.1)	282 (7.8)	2374 (65.9)
An. funestus	Dry	11 (0.31)	2 (0.06)	0	0	9 (0.25)	0	0	0	0	22 (0.6)
	Rainy	0	0	3 (0.02)	122 (3.4)	61 (1.7)	8 (0.4)	7 (0.2)	36 (1)	6 (0.2)	243 (6.8)
An. paludis	Dry	0	0	0	0	0	0	0	0	0	0
	Rainy	0	0	0	0	0	0	0	0	1 (0.03)	1 (0.03)
An. rufipes	Dry	16 (0.44)	0	0	1 (0.03)	0	0	0	0	0	17 (0.5)
	Rainy	1 (0.03)	0	0		46 (1.3)	0	0	0	0	47 (1.3)
An. ziemanni	Dry	0	0	0	0	0	0	0	0	0	0
	Rainy	0	0	1 (1.03)	33 (0.9)	18 (0.5)	0	0	0	0	52 (1.4)
An. pharoensis	Dry	7 (0.2)	0	0	2 (0.06)	0	0	0	4 (0.1)	0	13 (0.4)
	Rainy	0	1 (0.03)	11 (0.3)	26 (0.7)	5 (0.1)	0	0	0	0	43 (1.2)
An. coustani	Dry	0	0	0	0	0	0	0	0	0	0
	Rainy	0	0	0		10 (0.3)	3 (0.1)	0	0	55 (1.5)	68 (1.9)
An. squamasus	Dry	0	0	0		1 (0.03)	22 (0.6)	0	0	0	23 (0.64)
	Rainy	0	0	0	1 (0.03)	22 (0.6)	0	0	0	0	23 (0.64)
An. nili	Dry	0	0	0	0	1 (0.03)	0	0	0	0	1 (.03)
	Rainy	0	0	0	0	8 (0.2)	10 (0.3)	0	0	0	18 (0.5)
An. moucheti	Dry	0	0	0	0	0	0	0	0	0	0
	Rainy	0	0	0	0	1 (0.03)	0	0	0	0	1 (0.03)

Table 3
Circumsporozoite rate and entomological inoculation rate of anophelines in the different study sites in the dry and rainy season.

Localities	Godola CSP [95% CI] (EIR)	Mokolo CSP [95% CI] (EIR)	Djippordé CSP [95% CI] (EIR)	Gounougou CSP [95% CI] (EIR)	Mayo Mbocki CSP [95% CI] (EIR)	Vourgné Sanfou CSP [95% CI] (EIR)	Essong CSP [95% CI] (EIR)	Mangoum CSP [95% CI] (EIR)	Ngwatta CSP [95% CI] (EIR)
Dry season									
An. gambiae s.s.	0	0	0	0	0	0	(0.1)	(0.5)	0
Rainy season									
An. arabiensis	6.1 [2.4–9.8]	14.3 [0.5–23.8]	4.3 [0.6–8.0]	1.5 [0.3–3.7]	4.4 [1.2–7.5]	0	0	0	0
	(0.05)	(0.1)	(0.05)	(0.05)	(0.08)	0	0	0	0
An. gambiae s.s.	0	0	0	0	6.9 [1.6–12.2]	0	0.9 [0–1.7]	8.9 [6.1–11.7]	8.8 [4.9–12.7]
	0	0	0	0	(0.11)	0	(0.54)	(5.0)	(1.4)
An. funestus	0	0	0	0.8[0.4–1.2]	1.6[0.2–2.6]	0	0	0	0
	0	0	0	0.05	0.05	0	0	0	0
An. pharoensis	0	0	0	7.7[0.9–15.3]	0	0	0	0	0
	0	0	0	0.1	0	0	0	0	0
An. coustani	0	0	0	0	10[0.3–19.7]	0	0	0	0
	0	0	0	0	(0.05)	0	0	0	0

CSP = circumsporozoïte protein; CI = confidence intervals; EIR: entomological inoculation rate = number of infective bites per man per night.

et al., 2005; Robert et al., 1985). However, the number of *An. arabiensis* caught was by far greater than *An. funestus*. The occurrence of *An. arabiensis* during both seasons maybe due to the presence of stagnant water bodies such as marshes in and around the irrigated area or water from stand pipes or leaking from broken water pipes, stream-bed pools during the dry season and numerous puddles during the rainy season.

The mean biting rates for *An. funestus* peaked during the rainy season, and particularly in the irrigated rice paddies in Gounougou. The presence of this species was closely linked to rice cultivation, as most of its larvae were collected in the rice paddies, which is preferential for its breeding. Typical breeding sites for *An. funestus* consist of more-or-less permanent water bodies with emergent vegetation (Gillies and De Meillon, 1968); its scarcity throughout the dry season might have been the result of a halt in rice cultivation, owing to interruption in the irrigation system. Overall, higher anopheline densities were recorded in Gounougou and Mayo Mbocki compared to the much more urban villages of Mokolo and Djippordé, and the peri-urban village of Godola.

In the southern parts of the country, *An. gambiae* s.s. was the dominant vector species. These results confirm the confinement of the sibling *An. gambiae* s.s. to the forested areas of Africa (Coetzee et al., 2000; Wanji et al., 2003). *An. gambiae* s.s. was the main malaria vector in this area, and was found to be infected both in the dry and in the rainy seasons. This species is a well known and efficient vector of malaria in Cameroon and other areas of Africa and Madagascar (Meyus et al., 1962; Lindsay et al., 1990; Fondjo, 1996; Lengeler et al., 1997; Manga et al., 1997; Lindblade et al., 1999; Ijumba and Lindsay, 2001; Jambou et al., 2001; Keiser et al., 2004; Okollo et al., 2006; Bigoga et al., 2007) In Essong, malaria transmission was higher in the dry season than the rainy season compared to Mangoum. This might have been due to the presence of fish ponds, which boost the propagation of the vectors. In the rural Mangoum village, peak malaria transmission was recorded in the rainy season probably due to the availability of stagnant water bodies such as marshes around the farming areas and numerous puddles during the rainy season that constitute prolific breeding sources.

Although the use of ITN is a key intervention strategy for malaria control in Cameroon, its coverage rate is low (~25%) with the majority of its users in urban areas. However, the effectiveness of ITNs is not only determined by coverage, adherence, using nets properly, and periodic re-treatment of the nets, but also on the vector species involved in malaria transmission (Guillet et al., 2001). Scaling up coverage of ITNs is now the absolute priority for the control of malaria in Cameroon. From the observed transmission patterns in this survey, coupling the use of ITN with indoor residual spraying (IRS) would greatly impact on control, in response to the changing transmission patterns in the low transmission areas of northern Cameroon. Long-lasting insecticidal nets (LLINs) are now available and are the appropriate response for low re-treatment rates of conventional ITNs. Basic and continual education should be an integral part of the malaria control strategy to assure the use of INTs every night.

The findings of the present study highlight the paradigm of malaria transmission and delineate the role of anopheline species in malaria transmission in the nine sites. It complements existing data and would impact on the effective planning and implementation of evidence-based malaria vector intervention strategies in Cameroon and other areas with similar bio-geographical conditions.

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